

Answer Key — Biology Grade 11

Sub-Strand 1.4: Cell Division • Lesson 1: From Newborn to Teenager: Where Do All the New Cells Come From?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6
B	B	B	A	A	C

USE AFTER: PREDICT PHASE — Predict: 'Where did the extra 47 kg come from?' board question

Q1. A newborn baby in Kisumu weighs about 3 kg. By about 14 years the same child weighs 50 kg. How much body mass has the child gained?

- A. 17 kg
- B. 47 kg**
- C. 3 kg
- D. 53 kg

Correct answer: B. Mass gained = final mass - starting mass = 50 - 3 = 47 kg. Adding the masses gives 53 kg, and $50 \div 3$ is about 17, which is how many times heavier the teenager is, not the mass gained.

USE AFTER: OBSERVE PHASE — Observe: second viewing of the time-lapse, recording what changes (height, mass, face, limbs)

Q2. While watching a time-lapse of a child growing up over 14 years, the teacher says: 'Record only what you SEE, not what you think.' Which of these is something a learner SEES?

- A. Cells in the child's bones are multiplying
- B. The child becomes taller and heavier, and the face and limbs change**
- C. The child's cells are only getting bigger
- D. Food is turning directly into new body parts

Correct answer: B. Getting taller and heavier and changes in the face and limbs can be seen in the video. What is happening to cells inside the body cannot be seen, so the other choices are inferences.

USE AFTER: EXPLAIN PHASE — Explain: Discussion Diamond ranking of the statements (a) to (d)

Q3. A newborn has a mass of about 3 kg and a teenager has a mass of 50 kg or more. Which statement describes how the cells of the teenager's body compare with those of the newborn?

- A. The teenager has the same number of cells, and each cell is just much bigger
- B. The teenager's body is made of a very much larger number of cells than the newborn's body**
- C. Each baby is born with all the cells it will ever need, so the number never changes
- D. The teenager has fewer cells, but they are heavier

Correct answer: B. The lesson outcome is that growth from infancy to adolescence involves a very large increase in the number of cells. The other choices are the starting misconceptions of 'cells only get bigger' and 'babies are born with all their cells'.

USE AFTER: EXPLAIN PHASE — Explain: Gallery Walk of initial models and Discussion Diamonds

Q4. After the Gallery Walk, the teacher says: 'These are our starting models; we will test them.' What does this mean for the drawings the groups made?

- A. The drawings are first ideas that will be tested with evidence and improved in later lessons**
- B. The teacher already knows which drawing is correct and will mark the others wrong
- C. Only the drawings that look the same as most of the class will be kept
- D. The drawings are final and will not change for the rest of the unit

Correct answer: A. Initial models are a Version 1 baseline. They are revisited and revised as the class gathers evidence in later lessons, and none is marked right or wrong now.

USE AFTER: DQB PHASE — DQB: writing 2 questions on strips and posting under 'How does growth occur?'

Q5. Learners are writing questions for the Driving Question Board about how a 3 kg baby becomes a 50 kg teenager. Which question is the best one to post, because it can be investigated and goes beyond yes or no?

- A. How does a single cell become billions of cells?**
- B. Did the child in the video grow taller?
- C. Is growing good for you?
- D. Do babies drink milk?

Correct answer: A. A good DQB question starts with How, Why or What, links to the phenomenon, and can be investigated. The other choices are yes/no questions or are already answered by the video.

USE AFTER: END OF LESSON — Model: exit ticket and 'What I am not sure about' section of the initial model

Q6. At the end of Lesson 1, a class has watched a child grow and drawn initial models. Which of these questions can the class NOT yet answer with evidence from the lesson?

- A. About how many kilograms heavier is a 50 kg teenager than a 3 kg newborn?
- B. Did the child's height and appearance change in the time-lapse?
- C. Where do all the new cells come from, and how does each new cell get the right instructions?**
- D. Did the child's body mass increase between infancy and adolescence?

Correct answer: C. The class observed the changes and can work out the mass gained. Where the new cells come from and how each gets the right instructions is the puzzle left open for the unit, so learners should say 'we do not know yet'.